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September 9, 2024

Tensor product random matrix theory

I. REMARKS

- Reference: Alexander Altland, Joaquim Telles de Miranda, and Tobias Micklitz, *Tensor product random matrix theory*, arXiv:2404.10919 (2024). Note: last to authors are ordered reversively on the arXiv page.
- Alexander Altland will present this work at next week's workshop.
- Video of (probably the same) talk: `https://scgp.stonybrook.edu`.

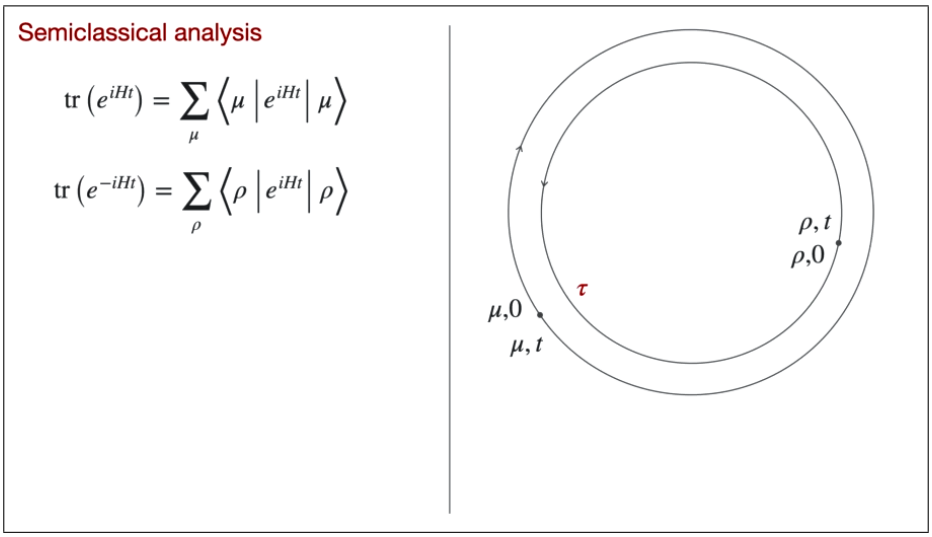
II. INTRODUCTION

Focus: spectral form factor

$$K(t) = \mathrm{Tr}(e^{iHt})\,\mathrm{Tr}(e^{-iHt}) \tag{1}$$

$$= \sum_{\mu,\nu} \langle\mu|e^{iHt}|\mu\rangle\langle\nu|e^{-iHt}|\nu\rangle. \tag{2}$$

To leading approximation, only paths $|\mu\rangle \to |\mu\rangle$ and $|\nu\rangle \to |\nu\rangle$ traversing the same sequence of scattering events in retarded and advanced chronology contribute (constructive interference). Result: $K(t) \sim t$ at early times. Known as the *diagonal approximation* of *Gutzwiller trace formula*.



Two non-interacting systems $H = H_A + H_B$:

$$K(t) = \sum_{\mu\nu,\rho\sigma} \langle\mu\nu|e^{i(H_A+H_B)t}|\mu\nu\rangle\langle\rho\sigma|e^{-i(H_A+H_B)t}|\rho\sigma\rangle. \tag{3}$$

Similar argument, but now paths need to match in both the Hilbert space of subsystem A and the Hilbert space of subsystem B. Result: $K(t) \sim t^2$ at early times.

Interacting system $H = H_A + H_B + H_{AB}$:

$$N_A\langle|H_{A\mu\mu'}|^2\rangle = N_B\langle|H_{A\nu\nu'}|^2\rangle = \lambda^2, \qquad N_A N_B\langle|H_{AB\mu\nu,\rho\sigma}|^2\rangle = \Lambda^2. \tag{4}$$

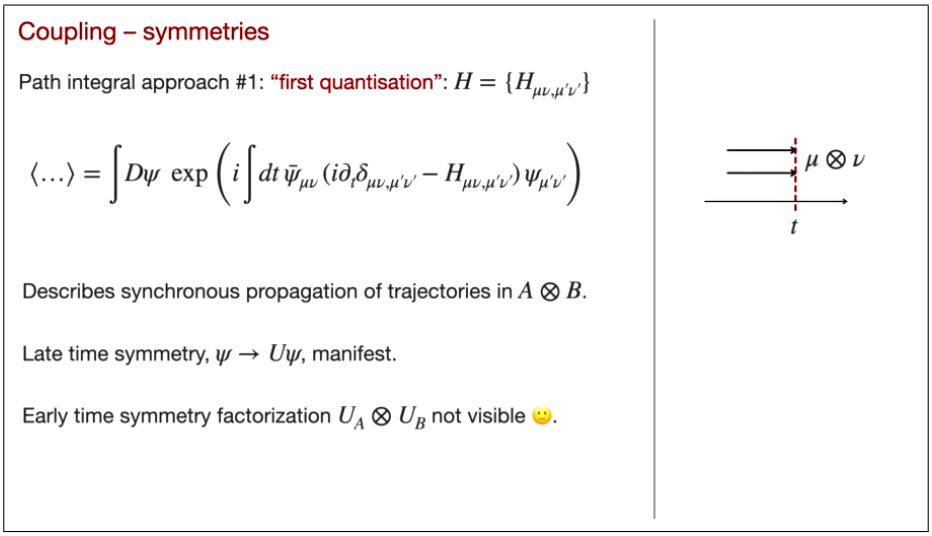
Fermi's golden rule: coupling rate $\gamma \sim N_A N_B \Lambda^2/2\lambda$.

Time $t \approx \gamma^{-1}$: crossover from $K(t) \sim t^2$ to $K(t) \sim t$. **Goal: describe the spectral form factor in the crossover regime.**

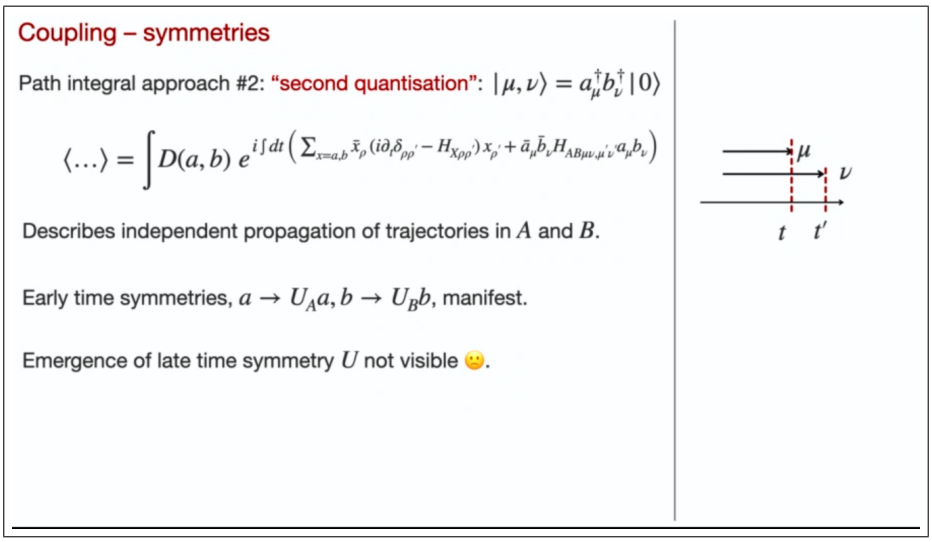
III. NAIVE FIELD THEORY APPROACHES

First quantization

A $(\dots)$ after the equation in the slide is missing. The late time symmetry is manifest as H is a (essentially, rotationally-invariant) random matrix. The $K(t) \sim t$ regime can be anticipated, the $K(t) \sim t^2$ not.



Second quantization



IV. MORE INVOLVED APPROACH

Disorder averaging (SYK model, arXiv:1711.08482):

$$Z(J_{ijkl}) = \int D\psi_i \exp\Big(- \int d\tau \Big[\frac{1}{2} \sum_i \psi_i \partial_\tau \psi_i + \sum_{1\le i<j<k<l\le N} J_{ijkl}\psi_i\psi_j\psi_k\psi_l \Big]\Big). \tag{5}$$

$$\langle Z\rangle_J \sim \int dJ_{ijkl} \exp\left(- \sum_{1\le i<j<k<l\le N} \frac{J_{ijkl}^2}{2\frac{3!J^2}{N^3}} \right) Z(J_{ijkl}). \tag{6}$$

Gaussian integration:

$$\langle Z\rangle_J \sim \int D\psi_i \exp\Big(- \int d\tau \frac{1}{2} \sum_i \psi_i \partial_\tau \psi_i \tag{7}$$

$$+ \sum_{1\le i<j<k<l\le N} \frac{3J^2}{N^3} \int\int d\tau d\tau' (\psi_i\psi_j\psi_k\psi_l)(\tau)(\psi_i\psi_j\psi_k\psi_l)(\tau')\Big). \tag{8}$$

Grassmann variables square to zero:

$$\sum_{1\le i<j<k<l\le N} (\psi_i\psi_j\psi_k\psi_l)(\tau)(\psi_i\psi_j\psi_k\psi_l)(\tau') = \frac{1}{4!}\Big[\sum_i \psi_i(\tau)\psi_i(\tau')\Big]^4. \tag{9}$$

GΣ-transformation:

$$1 = \int DG\delta\Big(NG(\tau,\tau') - \sum_i \psi_i(\tau)\psi_i(\tau')\Big) \tag{10}$$

$$\sim \int DGD\Sigma \exp\Big(- \frac{N}{2} \int\int d\tau d\tau' \Sigma(G - \frac{1}{N}\sum_i \psi_i\psi_i)\Big). \tag{11}$$

Σ plays the role of a Lagrange multiplier enforcing the delta constraint.

$$\langle Z\rangle_J \sim \int D\psi_i DGD\Sigma \exp\Big(- \int \frac{1}{2}\sum_i \psi_i\partial_\tau\psi_i - \frac{1}{2}\int\int N\Sigma(G - \frac{1}{N}\sum_i \psi_i\psi_i) \tag{12}$$

$$+ \frac{J^2 N}{2\cdot 4}\int\int G^4\Big). \tag{13}$$

Exponential is bilinear in the fermion variables ψ_i.

$$\langle Z\rangle_J \sim \int DGD\Sigma \Big[\det(\partial_\tau - \Sigma)\Big]^{\frac{N}{2}} \exp\Big(- \frac{N}{2}\int\int (\Sigma G - \frac{1}{4}J^2G^4)\Big) \tag{14}$$

$$= \int DGD\Sigma e^{-NI[G,\Sigma]}, \tag{15}$$

with

$$I[G,\Sigma] = -\frac{1}{2}\log\det(\partial_\tau - \Sigma) + \frac{1}{2}\int\int (\Sigma G - \frac{1}{4}J^2G^4). \tag{16}$$

Upshot: N in front of the action in the second line of (14). The classical equations of motion can be derived by looking for extrema of this action.

Back to the original problem:

$$S_X[G,\Sigma] = N_X \operatorname{Tr}\Big(\ln(i\partial_t - \Sigma_X) + G_X\Sigma_X + \frac{\lambda^2}{2}G_X^2\Big) \tag{17}$$

$$S_{AB}[G] = \frac{N_A N_B}{2\Lambda^2}\int dt \int dt' \operatorname{Tr}((G_A \odot G_B\tau_3)_{tt'}(G_A\odot G_B\tau_3)_{t't}) \tag{18}$$

Here, $\odot$ denotes an element-wise product.

Stationary phase approach:

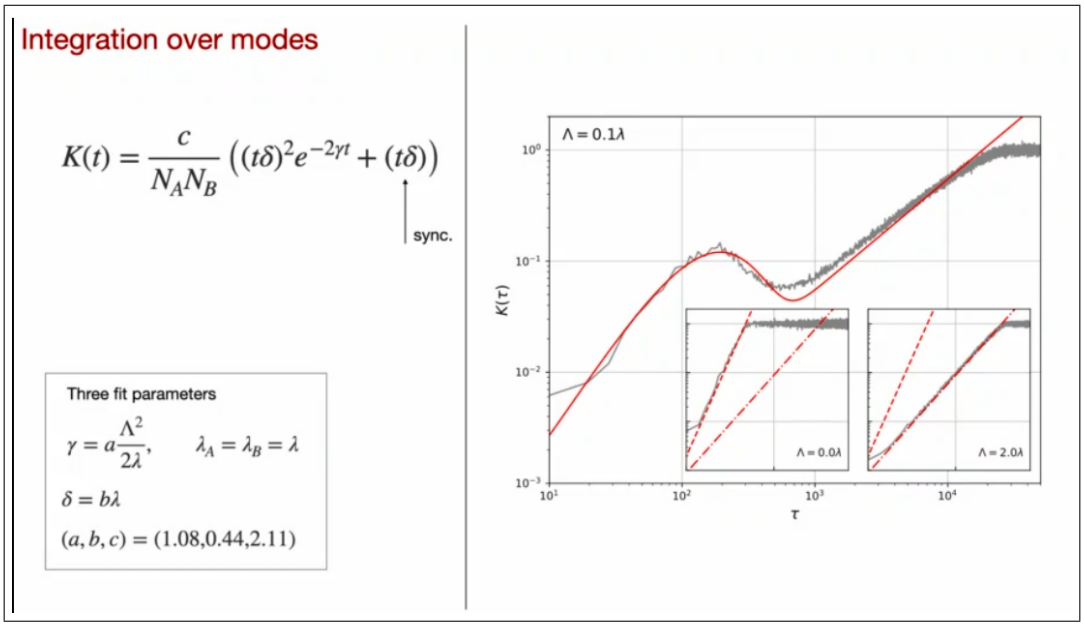
Ignoring the interaction for the moment, variation of the action leads to

$$G_X = \frac{1}{i\partial_t - \Sigma_X}, \qquad \Sigma_X = \lambda^2 G_X. \tag{19}$$

Low frequencies: $\partial_t \ll \Gamma_X$, leading to a rotational symmetry:

$$G_X = -iT_X \tau_3 T_X^{-1} \equiv -iQ_X \tag{20}$$

Crucial observation: rotational symmetry does not lock the fields in $S_{AB}[G]$. Leads to $K(t) \sim t^2$.

Late times: ∂_t can no longer be ignored, perturbation theory.

V. FINAL RESULT